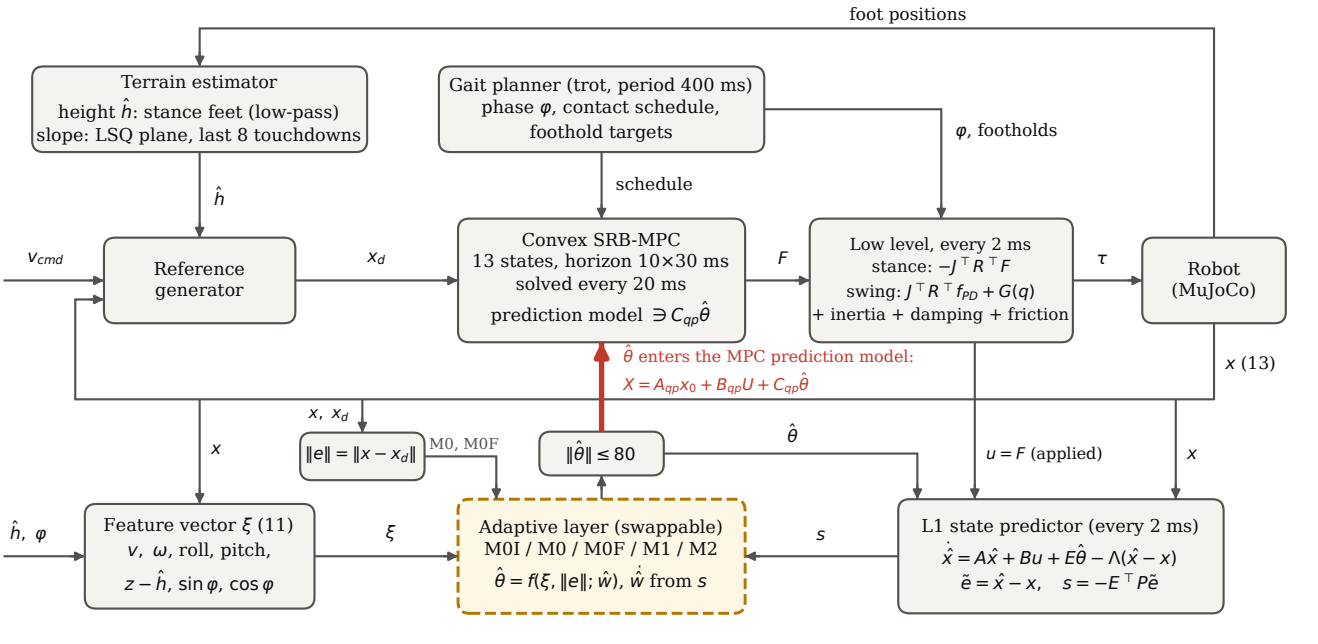
foot positions
Terrain estimator
height $\hat{h}$: stance feet (low-pass)
slope: LSQ plane, last 8 touchdowns
Gait planner (trot, period 400 ms)
phase φ, contact schedule,
foothold targets
φ, footholds
schedule
$\hat{h}$
v_{cmd}
Reference generator
x_d
Convex SRB-MPC
13 states, horizon 10×30 ms
solved every 20 ms
prediction model $\ni C_{qp}\hat{\theta}$
F
Low level, every 2 ms
stance: $-J^\top R^\top F$
swing: $J^\top R^\top f_{PD} + G(q)$
+ inertia + damping + friction
τ
Robot (MuJoCo)
x (13)
$\hat{\theta}$ enters the MPC prediction model:
$X = A_{qp}x_0 + B_{qp}U + C_{qp}\hat{\theta}$
x, x_d
$\|e\| = \|x - x_d\|$
M0, M0F
$\|\hat{\theta}\| \le 80$
$\hat{\theta}$
$u = F$ (applied)
x
$\hat{h}$, φ
Feature vector ξ (11)
v, ω, roll, pitch,
$z - \hat{h}$, $\sin\varphi$, $\cos\varphi$
ξ
Adaptive layer (swappable)
M0I / M0 / M0F / M1 / M2
$\hat{\theta} = f(\xi, \|e\|; \hat{w})$, $\dot{\hat{w}}$ from s
s
L1 state predictor (every 2 ms)
$\dot{\hat{x}} = A\hat{x} + Bu + E\hat{\theta} - \Lambda(\hat{x} - x)$
$\tilde{e} = \hat{x} - x$, $s = -E^\top P\tilde{e}$